Lecture 1: Introduction to Imaging Science in Medicine

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Common Jargon

Subject: Entity to be imaged, e.g., human, mouse, etc.

Object f: Distribution of physical properties that the imaging system seeks to estimate

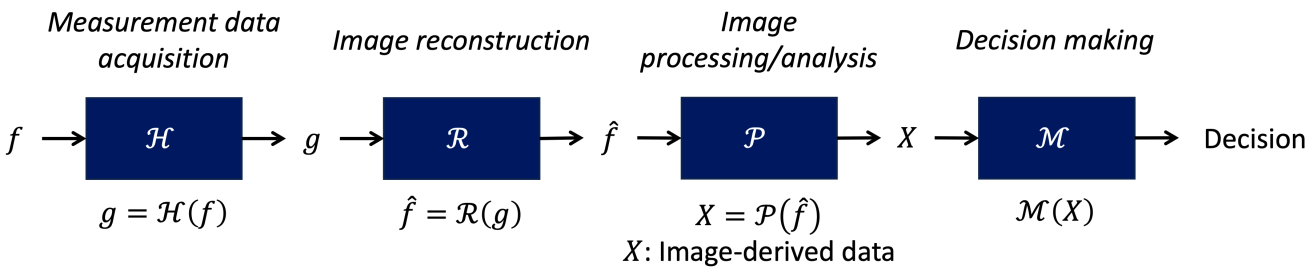
Measured image data g: Physical measurements acquired by the imaging system

Image $\hat{f}$: Representation of the form of an object

Imaging Science

Multidisciplinary field, where computational algorithms and techniques are developed to overcome the limitations of traditional imaging systems and to extract information from the measured data

Typical Pipeline of Imaging in Medicine



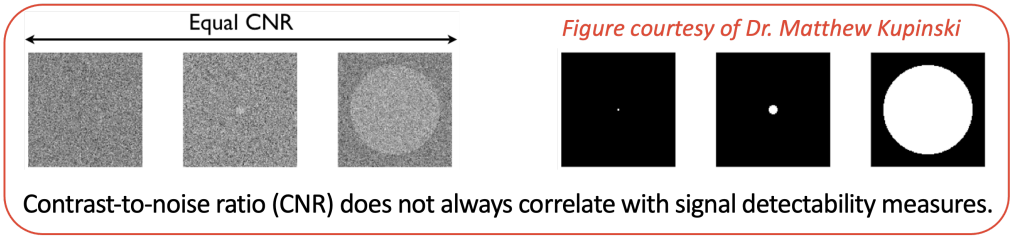
Optimization of imaging systems and related technology ➡️ Better decision

Let's assume we want to evaluate the quality of a reconstructed or processed image to make better decisions.

❓ **How would you define an image of good quality?**

❓ **What kind of measure would you use to assess the image quality?**

Example of traditional physical measure



Images should be **informative with respect to specific tasks**, such as signal detection, *rather than simply "good-looking."*

➡️ **Objective Assessment of Image Quality (OAIQ)**

Example: Process of Approving New Medical Imaging Technologies

